Assignment 01

1. Mass Customization is:
A. A same product available for large nnumber of people.
B. A modular product that can be reconfigured as needed.
C. A few people using customized product.
D. Making a rare product.

2. Which of following is not a type of product design?
A. Design by evolution
B. Design by innovation
C. Design by disintegration
D. All of these

3. Design by innovation happens as:
A. Minor changes in design for improving performance.
B. Rapid changes in a short time period with scientific discovery.
C. Small changes over a prolonged period.
D. An evolved design towards mass production.

4. Optical surgery is an example of:
A. Design by Intuition
B. Design by Evolution
C. Design by Innovation
D. Design by Integration

5. _____ is the backbone of engineering designs.
A. Ideation.
B. Information Data collection & redefinition.
C. Socio-economic status & values
D. Product realization.

6. Ideally, data should not be:
A. Stored logically after its use.
B. Destroyed just for being confidential.
C. Able to be retrieved for future use.
D. Encrypted and stored at a third party.

7. Which of the following is not a common manufacturing practice.
A. Reduction of number of parts and assembly.
B. Standard practice like standard part & tolerances.
C. Ensuring testability and reparability by using pre-built test methods.
D. Use of large variety of material.

8. Which of the following is not an essence of S curve of technology growth cycle?
A. Continual Innovative design process.
B. Large number of early adaptors in customer pool.
C. Reconfigurable manufacturing facility.
D. Static Technology.

9. Design for Manufacturing may not emphasize:
A. Socio-economic factors.
B. Assembly of parts.
C. Maintenance.
D. It emphasizes on all of these

10. Formulation in Problem Solving process refers to:
A. Using various mode of communication to know the problem.
B. Well documentation of the problem so that it can be retrieved when needed.
C. Turning crude and qualitative problems into quantitative measures.
D. Iterating on problem statement for Ideation.

Assignment 01

Q1. Product is not referred as:

A. Merchandise
B. Finished Good
C. Service
D. Divinity

Q2. Mass production happens only after:

A. Minimum viable product is ready
B. Testing a product for its full service period
C. Absolutely leaving no place for improvements
D. Company test with sample size customer and found product satisfactory

Q3. Design by evolution happens as:

A. Skeletal design is intact and features are added or removed.
B. Design as a whole requires radical change
C. Significant changes are rquired in a short time period
D. Existing design is good enough and doesn't need a change

Q4. Development of digital camera from a reel based camera is example of:

A. Design by Intuitions
B. Design by Evolution
C. Design by Innovation
D. Design by Integration

Q5. Which of the following is not a step of Product-Consumption Cycle:

A. Production
B. Distribution
C. Evolution
D. Consumption

Q6. A designer should not:

A. Quantify the customer's need
B. Be monotonous & unidirectional
C. Take account of the end life of a product
D. Consider economic perspective

Q7. Customer satisfaction may not be approximated by:

A. Reliability

B. Repeatability
C. Durability
D. Versatility

Q8. Sales versus Profit plot is:

A. Linear from start and sudden drop at end life
B. Maximum in maturity period
C. Negative in maturity period
D. Always positive in initial phase

Q9. In concurrent engineering, a designer will not have to deal with:

A. Incurring cost
B. Latest technology
C. Interdisciplinary aspects
D. None of these

Q10. Engineering Problem Solving involves process in following order:

A. Analysis > Formulation of problem > Review of alternatives > Comparing alternatives > Specifying best solution.
B. Formulation of problem > Analysis > Review of alternatives > Comparing alternatives > Specifying best solution.
C. Formulation of problem > Analysis > Comparing alternatives > Review of alternatives > Specifying best solution.
D. Formulation of problem > Comparing alternatives > Review of alternatives > Analysis > Specifying best solution.

Assignment 02

Q1. Which of the following is important while designing a product?

A. Industrial design
B. Mechanical design
C. Both of these
D. None of these

Q2. Product characteristics curve is plotted with respect to:

A. Revenue vs time
B. Revenue vs production
C. Production vs time
D. Production vs consumption

Q3. Design is a cyclic process

A. True
B. False

Q4. Which of he following factor is not primarily considered while setting initial product objectives:

A. Performance
B. Environment
C. Target product cost
D. Testing

Q5. Which of the following is a factor of product quality?

A. Portability
B. Cost
C. Durability
D. All of these

Q6. First phase of the morphology of design is:

A. Detailed design
B. Feasibility study
C. Planning for consumption
D. Planning for retirement

Q7. The sequence of the phases of morphology of design is:

A. Preliminary design, Detailed design, Feasibility study, Production process, Planning for distribution, Planning for use, Planning for Retirement.

B. Planning for use, Planning for Retirement, Preliminary design, Detailed design, Feasibility study, Production process, Planning for distribution.
C. Detailed design, Feasibility study, Preliminary design, Production process, Planning for distribution, Planning for use, Planning for Retirement.
D. Feasibility study, Preliminary design, Detailed design, Production process, Planning for distribution, Planning for use, Planning for Retirement.

Q 8. Dependability is the length of the service life or endurance of a product under given working conditions.

A. True
B. False

Q 9. Operational aspects do not include:

A. Fundamental reason for the product demand
B. Durability
C. Dependability
D. All of these

Q10. Which of the following is attribute of successful product development?

A. Cost
B. Value
C. Time
D. All of these

Assignment 02

Q 1. Promotional activity is planned in which phase:

A. Designing phase
B. Planning for production
C. Planning for distribution
D. Planning for consumption

Q 2. Match the following:

(a) Direct analogy	(1) Correlating with paradoxical phrase
(b) Personal analogy	(2) Correlating with imagination
(c) Symbolic analogy	(3) Correlating with the situation from a different field
(d) Fantasy analogy	(4) Correlating with the self-experience

Select the correct option

A. a-1, b-2, c-3 and d-4
B. a-2, b-3, c-4 and d-1
C. a-3, b-4, c-1 and d-2
D. a-4, b-3, c-2 and d-1

Q 3. Product quality is not significantly defined by:

A. Durability
B. Dependability
C. Cost
D. All of these

Q 4. Choose the correct sequence of the morphology of design:

A. Feasibility study, Preliminary design, Detailed design, Production process, Distribution, Consumption, Retirement
B. Preliminary design, Detailed design, Feasibility study, Production process, Distribution, Consumption, Retirement
C. Consumption, Retirement, Preliminary design, Detailed design, Feasibility study, Production process, Distribution,
D. Detailed design, Feasibility study, Preliminary design, Production process, Distribution, Consumption, Retirement

Q 5. The last phase of the morphology of design is:

A. Detailed design

B. Feasibility study
C. Planning for consumption
D. Planning for retirement

Q 6. Attributes of successful product development are cost, value, time and technical know-how.

A. True
B. False

Q 7. Product is characterized in terms of:

A. Functional or performance aspects
B. Aesthetic or appearance aspects
C. Operational or ease of use
D. All of these

Q 8. Functional aspect of product characteristics implies:

A. Fundamental reason for the product demand
B. How the product operates/works while in use
C. Both of these
D. None of these

Q 9. Durability is the length of the service life or endurance of a product under given working conditions.

A. True
B. False

Q 10. Operational aspects include:

A. Fundamental reason for the product demand
B. Ease to handle and operate
C. Durability and dependability
D. None of these

Assignment 03

Q1. Match the following:

(a) Regularly assemblages of repeated or other varied elements	(1) Repetition
(b) Process of creating an identical instance of an element	(2) Variation
(c) Process of creating non-identical instance of an element	(3) Patterning

Select the correct option

A. a – 1, b-2 and c-3
B. a – 2, b-3 and c-1
C. a – 3, b-1 and c-2
D. None of these

Q2. Fractal is a shape that is self-similar and has a fractional dimension.

A. True
B. False

Q3. What is the full form of QFD:

A. Quality Function Deployment
B. Quality Fitted Data
C. Quality Function Data
D. None of these

Q4. Yellowish grey is called:

A. Russet
B. Buff
C. Citron
D. Olive

Q5. Choose the correct sequence of QFD/HOQ phases:

A. Design Characteristics, Process, Product/Services, Control
B. Design Characteristics, Product/Services, Process, Control
C. Control, Design Characteristics, Process, Product/Services
D. Design Characteristics, Product/Services, Control, Process

Q6. Which of the following is a primary color?

A. Orange
B. Violet
C. Blue

D. Orange

Q7. Quality of color is not controlled by:

A. Hue
B. Intensity
C. Value
D. Radiance

Q8. Number that represents the golden ratio is:

A. 1.8610
B. 1.1680
C. 1.6180
D. 1.0618

Q9. Elements of design are:

A. Shape
B. Texture
C. Color
D. All of these

Q10. The Kano Model is a tool that is used to identify customer needs.

A. True
B. False

Assignment 03

Q1. In visual design:

A. Imagination is followed
B. Visual communication principles are followed.
C. Both of these
D. None of these

Q2. Quality of color is controlled by:

A. Hue
B. Intensity
C. Value
D. All of these

Q3. Which of the following is not a primary color?

A. Yellow
B. Red
C. Blue
D. Green

Q4. What is/are the elements of design:

A. Form
B. Texture
C. Color
D. All of these

Q5. Type of product evolutions is/are:

A. Technology push
B. Markets pull
C. Both of these
D. None of these

Q6. Rhythm in design is created by:

A. Color
B. Music
C. Both of these
D. None of these

Q7. Guidelines for design include:

A. Rhythm
B. Emphasis
C. Balance
D. All of these

Q8. Match the following:

(a) Hue with added white	(1) Shade
(b) Hue with added black	(2) Tint
(c) Hue with added complement	(3) Tone

Select the correct option

A. a-1, b-2 and c-3
B. a-1, b-3 and c-2
C. a-2, b-3 and c-1
D. None of these

Q9. Basic principle of form/shape is given by:

A. Gestalt
B. Newton
C. Archimedes
D. Galileo

Q10. Emphasis is achieved through:

A. Color
B. Shape
C. Size
D. All of these

Assignment 04

Q1. Choose the correct sequence of the phases of Value Engineering methodology:

A. Information, Creativity, Function analysis, Evaluation, Development, Implementation
B. Information, Creativity, Evaluation, Function analysis, Development, Implementation
C. Information, Function analysis, Creativity, Evaluation, Development, Implementation
D. Information, Creativity, Evaluation, Development, Function analysis, Implementation

Q2. Information phase does not include:

A. Marketing
B. Data collection
C. Team work
D. Finalization of problem

Q3. Match the following steps of the Function Analysis System Technique (FAST):

(a) Step 1	(1) Verify the FAST diagram
(b) Step 2	(2) Determine the highest order function
(c) Step 3	(3) Expand the FAST diagram
(d) Step 4	(4) Identify the basic functions
(e) Step 5	(5) Identify the supporting functions

Select the correct option

A. a – 1, b – 2, c – 3, d – 4 and e – 5
B. a – 2, b – 3, c – 4, d – 5 and e – 1
C. a – 1, b – 4, c – 3, d – 5 and e – 2
D. a – 2, b – 4, c – 3, d – 5 and e – 1

Q4. There exists a single unique/correct FAST diagram.

A. True
B. False

Q5. INVEST was established in:

A. October 1950
B. August 1977
C. October 1977
D. August 1950

Q6. Who is called father of Value Engineering?

A. Newton
B. Edwin B. Flippo
C. Archimedes
D. Lawrence D. Miles

Q7. In Value Engineering, value is defined as:

A. $\frac{Product\ utility}{Product\ cost}$

B. $\frac{Product\ cost}{Product\ utility}$

C. $\frac{Product\ price}{Product\ cost}$

D. $\frac{Product\ cost}{Product\ price}$

Q8. Which of the following is a direct advantage of value analysis?

A. Worker satisfaction
B. Reduction of scrap
C. Increase in productivity
D. All of these

Q9. In Value Analysis, the value is denoted by ratio of cost to function.

A. True
B. False

Q10. Economic value comprises of:

A. Use value
B. Esteem value
C. Exchange value
D. All of these

Assignment 04

Q1. Term ‘Value Analysis’ is first used in:

A. 1960
B. 1930
C. 1954
D. 1947

Q2. Technique to identify and eliminate unnecessary costs in design, testing, manufacturing, construction, operations, maintenance, procedure, specification, and practices while focusing on function of the product is known as:

A. Cost Engineering
B. Profit Engineering
C. Revenue Engineering
D. Value Engineering

Q3. Value engineering improves:

A. Value
B. Performance
C. Time
D. All of these

Q4. Value analysis is synonym to value engineering:

A. True
B. False

Q5. Who is the founder of ‘Value Engineering’ concept?

A. Lawrence D. Miles
B. Newton
C. Gestalt
D. Archimedes

Q6. Difference between value analysis and value engineering is:

A. Marketing
B. Implementation
C. Recovery
D. None of these

Q7. Term ‘Value Engineering’ is first used in:

A. 1960
B. 1930
C. 1954
D. 1947

Q8. Full form of ‘FAST’ is:

A. Function Analysis Syndrome Technique
B. Future Analysis System Technique
C. Future Analysis Syndrome Technique
D. Function Analysis System Technique

Q9. Value engineering is needed to deal with the:

A. Lagging productivity
B. Changing customer expectations
C. Both of these
D. None of these

Q10. Cost reduction is:

A. Item oriented
B. Function oriented
C. Both of these
D. None of these

Assignment 05

Q1. T300 is subclass _____ class of metals.

A. Al Alloy
B. Steels
C. Cu Alloy
D. Ti Alloy

Q2. Which is/are classified as secondary manufacturing process?

A. Machining
B. Turning
C. Grinding
D. All of these

Q3. Operational cost does not include.

A. Direct labor cost
B. Material cost
C. Overhead cost
D. None of these

Q4. Which of these is/are non-ferrous metal(s)?

A. Zinc
B. Iron
C. Steel
D. Glass

Q5. Which process diminishes the candidature of materials when their attributes lies outside the limits imposed by the design?

A. Screening
B. Cost optimization
C. Both of these
D. None of these

Q6. Which of the following is/are the mechanical property of robust product material?

A. Strength
B. Modulus
C. Fatigue
D. All of these

Q7. _____ is a physical property of material.

A. Oxidation
B. Melting point
C. Corrosion
D. Surface treatment

Q8 Machining process generally is not recommended for copper.

A. True
B. False

Q9. Which of the following is/are factor(s) of physical form?

A. Size
B. Weight
C. Shape
D. All of these

Q10. Ceramic comes under Ferrous metals.

A. True
B. False

Assignment 05

Q1. Design recommendations include:

A. Minimize material cost
B. Use of standard shapes
C. Both of these
D. None of these

Q2. Which is/are comes under ferrous metals:

A. Copper
B. Wood
C. Paper
D. Steel

Q3. Polymers comes under:

A. Ferrous metal
B. Non-ferrous metal
C. Inorganic material
D. Organic material

Q4. Ceramic comes under:

A. Ferrous metal
B. Non-ferrous metal
C. Inorganic material
D. Organic material

Q5. Which material is used for high temperature (2000˚C) application?

A. Epoxy
B. Lead
C. Carbon
D. Nylon

Q6. Which material can be re-melted and reused:

A. Thermoplastics
B. Thermosets
C. Both of these
D. None of these

Q7. Strength, modulus and fatigue classify under:

A. Mechanical factor
B. Processing and fabrication
C. Physical form
D. Life of the component

Q8. Which process is not recommended for copper?

A. Extrusion
B. Machining
C. Press forming
D. All of these

Q9. Choose the correct sequence of the selection of manufacturing process:

A. Identify processes, Selection criteria, Evaluate process, Selection
B. Selection criteria, Identify processes, Evaluate process, Selection
C. Identify processes, Evaluate process, Selection criteria, Selection
D. Selection, Identify processes, Evaluate process, Selection criteria

Q10. Manufacturing processes are evaluated on basis of:

A. Process capability
B. Degree of automation
C. Required post processing
D. All of these

Assignment 06

Q1. Recycling is majorly done to:

A. Reuse material only
B. Restore product to the working condition
C. Improve the product to the quality level though not like new
D. Restore to quality level as new

Q2. Methods used for evaluating the 'Design for Assembly' are:

A. Lucas DFA method
B. Methods Time Measurement Standards
C. Hitachi assembly evaluation method
D. All of these

Q3. Corrective maintenance is:

A. Reactive in nature
B. Proactive in nature
C. Predictive in nature
D. None of these

Q4. Maintenance location can be:

A. Stationary
B. Mobile
C. Remote
D. All of these

Q5. Preventive maintenance is:

A. Reactive in nature
B. Corrective in nature
C. Predictive in nature
D. None of these

Q6. Which of the following is/are the purpose of Design Review?

A. Provide systematic product-process analysis
B. Formal record of product-process analysis
C. Feedback to design team for improvement
D. All of these

Q7. Design review is a cyclic process.

A. True
B. False

Q8. ‘Design for Manufacturing’ does not include:

A. Optimization of all manufacturing function
B. Minimization of complexity of manufacturing process
C. Assuring the best
D. All of these

Q9. _____ is preferred for large volume production.

A. Manual assembly
B. One/two robotic arm assembly
C. Multi-station robotic assembly
D. Automatic assembly

Q10. Repair is mostly done to restore to quality level of a product as new.

A. True
B. False

Assignment 06

Q1. DFM focuses on:

A. Minimizing the complexities involved in manufacturing operations
B. Reducing the overall part production cost
C. Both of these
D. None of these

Q2. Design review is a:

A. One time process
B. Cyclic process
C. Both of these
D. None of these

Q3. Use of standard commercially available component in DFM is:

A. Not preferred
B. Does not matter
C. Preferred
D. Out of the scope

Q4. Is it compulsory to follow the same sequence of assembly in disassembly process?

A. Yes
B. No
C. Can't say
D. None of these

Q5. Which type of the assembly is preferred for large volume production?

A. Manual Assembly
B. One/two robotic arm assembly
C. Multi-station robotic assembly
D. Automatic assembly

Q6. DFA focuses on:

A. Reduction and standardization of parts and subassemblies and assemblies
B. Reducing or eliminating expensive, complex or unnecessary features
C. Reducing or eliminating labor
D. Reducing or eliminating inventory

Q7. What is a full form of MTM standards?

A. Mean Time Management Standards
B. Methods Time Management Standards
C. Methods Time Measurement Standards

D. Mean Time Measurement Standards

Q8. Which of the following methods is more internationally recognized?

A. Lucas DFA method
B. Methods Time Measurement Standards
C. Hitachi assembly evaluation method
D. None of these

Q9. Selective disassembly generally is:

A. Reversible dismantling
B. Irreversible dismantling
C. Both of these
D. None of these

Q10. Repair is done to:

A. Reuse material only
B. Restore product to the working conditions
C. Improve the product to the quality level though not like new
D. Restore to quality level as new

Assignment 07

Q1. What is full form of SPC:

A. Sample Production Control
B. Statistical Production Control
C. Statistical Process Control
D. Simple Process Control

Q2. Which factor least demand for Design for Environment?

A. Profit
B. Customer demand
C. Government pressure
D. ISO requirements

Q3. Full form of OCAP is:

A. Out-of-Control-Action Plans
B. On-the-Control-Action Plans
C. Over-the-Control-Action Plans
D. None of these

Q4. Full form of EPC is:

A. Excess Production Control
B. Excess Process Control
C. Engineering Production Control
D. Engineering Process Control

Q5. Process target value (T) is:

A. (USL*LSL)/2
B. (USL/LSL)/2
C. (USL+LSL)/2
D. (USL-LSL)/2

Q6. Which of the following is not a non- conventional energy source?

A. Wind
B. Fossil fuel
C. Solar
D. Tidal

Q7. _____ is not a predominantly used tool of SPC..

A. Histogram
B. Gantt chart
C. Pareto chart
D. Control chart

Q8. CERES stands for Coalition of Environmental Repository Economies.

A. True
B. False

Q9. _____ is not a term in the basic 3R.

A. Resize
B. Renew
C. Recycle
D. Reduce

Q10. Principles of Design for Manufacturing include:

A. Process
B. Design
C. Testing
D. All of these

Assignment 07

Q1. Non- conventional energy sources are:

A. Wind
B. Solar
C. Tidal
D. All of these

Q2. Fossil fuel is now a non-conventional energy source.

A. True
B. False

Q3. Full form of the CERES is:

A. Center of Environmental Responsible Economies
B. Coalition of Environmental Responsible Economies
C. Coalition of Environmental Repository Economies
D. Center of Environmental Repository Economies

Q4. Environmental impact is:

A. Global
B. Local
C. Both of these
D. None of these

Q5. Life cycle stages include:

A. Reuse
B. Composting
C. Disposition
D. All of these

Q6. The new term introduced in 4R's is:

A. Renewable energy
B. Reduce
C. Reuse
D. Recycle

Q7. What act as an input in LCA?

A. Land
B. Energy
C. Raw material
D. All of these

Q8. Each of the seven major activities in manufacturing sequence have the same potential to generate the waste.

A. True
B. False

Q9. The quality of a product can be decided by:

A. Performance
B. Durability
C. Reliability
D. All of these

Q10. Statistical methods for quality control and improvement include:

A. Design of experiments
B. Acceptance sampling
C. Both of these
D. None of these

Assignment 08

Q1. Buzz session technique is used for:

A. Analysis
B. Perception
C. Creativity
D. Assessment

Q2. The method of pooling large number of expert judgements through a series of increasingly refined questionnaires is known as:

A. Brainstorming technique
B. Morphological analysis
C. Attribute listing technique
D. Delphi technique

Q3. Creativity is maximum generally at _____ years of age.

A. 4-5
B. 10-12
C. 16-18
D. 25-27

Q4. Process Patent is also known as:

A. Utility Patent
B. Design Patent
C. Plant Patent
D. None of these

Q5. Frugal innovation is also known as:

A. Competitive Innovation
B. Sustainable Innovation
C. Regular Innovation
D. None of these

Q6. Provisional patent idea cannot be taken to provisionally hold the idea.

A. True
B. False

Q7. _____ aspect of the invention is patented under Utility Patent.

A. Functional
B. Maintenance
C. Management
D. None of these

Q8. The patent application in India is published after 18 months of first filing.

A. True
B. False

Q9. RFE stands for:

A. Reference for Examiner
B. Request for Examination
C. Request for Extension
D. Reply for External

Q10. _____ is general approach to problem solving.

A. Analytical approach
B. Creative approach
C. Both of these
D. None of these

Assignment 08

Q1. Functional aspect of the invention is patented under:

A. Design Patent
B. Plant Patent
C. Utility Patent
D. None of these

Q2. The patent application is published after _____ months of first filing.

A. 7
B. 14
C. 18
D. 20

Q3. Full form of RFE is:

A. Reference For Examiner
B. Request For Extension
C. Request For Examination
D. Reply For External

Q4. There are ____ steps for patent registration.

A. 5
B. 10
C. 11
D. 21

Q5. Provisional patent idea can be taken to provisionally hold the idea.

A. True
B. False

Q6. Analysis of function by using creativity is called the root of value engineering.

A. True
B. False

Q7. The general approach(es) to problem solving is/are:

A. Analytical approach
B. Creative approach
C. Both of these
D. None of these

Q8. What does not act as block for creativity?

A. Emotion

B. Culture
C. Flexibility
D. Habitual

Q9. Which factor is not conducive to creativity?

A. Perception
B. Originality
C. Observation
D. Idea Fluency

Q10. There are majorly _____ types of patents.

A. 2
B. 3
C. 4
D. 5

Assignment 09

Q1. What is the full form of AM fabrication?

A. Additive Machining fabrication
B. Additional Machining fabrication
C. Additive Manufacturing fabrication
D. Additional Manufacturing fabrication

Q2. Which kind of material is used in Rapid Prototyping?

A. Liquid
B. Solid
C. Discrete particles
D. All of these

Q3. What is the full form of SL?

A. Stereo Lithography
B. Straight Lithography
C. Stereo Light
D. Straight Light

Q4. What is the full form of SLS?

A. Stereo Laser Sintering
B. Selective Laser Sintering
C. Selective Lithography Sintering
D. Straight Light Sintering

Q5. Parts are made by _____ in Additive Manufacturing.

A. cutting material in layers
B. extracting material in layers
C. removing material in layers
D. adding material in layers

Q6. Layer technique was proposed first by _____.

A. Brubaker
B. Blanther
C. Walter Dandy
D. None of these

Q7. Which of the following material can be used for Additive Manufacturing?

A. Polymers
B. Metals
C. Ceramics
D. All of these

Q8. _____ is the most recent technology.

A. Manual Prototyping
B. Automated Prototyping
C. Digital Twins
D. Rapid Prototyping

Q9. The first step of rapid prototyping is _____.

A. Costing
B. 3D modeling
C. Data conversion
D. Production

Q10. Factory cost is: Material Cost + Operation Cost - Labor Cost + Factory Overhead.

A. True
B. False

Assignment 09

Q1. Product design and development is rapid but the manufacturing is traditional in:

A. Traditional manufacturing
B. Additive Manufacturing
C. Rapid Prototyping
D. None of these

Q2. All the products can be manufactured using Additive Manufacturing:

A. True
B. False

Q3. Digital manufacturing is included in:

A. Traditional manufacturing
B. Rapid Prototyping
C. Additive Manufacturing
D. None of these

Q4. Layer technique was proposed by Blanther in:

A. 1908
B. 1809
C. 1980
D. 1890

Q5. Photo sculpture was introduced in.

A. Nineteenth century
B. Twelfth century
C. Eighth century
D. None of these

Q6. Rapid Prototyping is _____ of Concurrent Engineering.

A. independent
B. an integral part
C. dependent part
D. None of these

Q7. Concurrent Engineering is also known as:

A. Total compression engineering
B. Time compression engineering
C. Time compression evaluation
D. Time cumulative engineering

Q8. What is/are the display system to make 3D outlines?

A. Wireframe model
B. Surface model
C. Solid model
D. All of these

Q9. Raw material that can be used in ‘Additive Manufacturing’ should be in:

A. Solid block form
B. Liquid form
C. Gaseous form
D. All of these

Q10. Product design, development and manufacturing is rapid in:

A. Traditional manufacturing
B. Additive Manufacturing
C. Rapid Prototyping
D. None of these

Assignment 10

Q1. Brim is generally removed in polymer FDM by:

A. Scissor
B. Hammer
C. Chisel
D. Hand

Q2. Unit production cost for mass manufacturing is low in:

A. Functional layout
B. Line layout
C. Cellular layout
D. Combination layout

Q3. Bed of the demonstrated machine can move in:

A. Y direction
B. Z direction
C. X direction
D. None of these

Q4. Pareto has a _____ rule.

A. 90-10
B. 20-80
C. 60-40
D. 40-60

Q5. Minimal material movement is observed in:

A. Fixed layout
B. Functional layout
C. Cellular layout
D. Combination layout

Q6. Function layout is also known as:

A. Product layout
B. Cellular layout
C. Process layout
D. Combination layout

Q7. Which of the following is an advantage of Process layout?

A. Flexibility
B. System protection
C. Less interruptions
D. All of these

Q8. Which of the following is true about filament in FDM?

A. It is final product after printing
B. It is raw material used for printing
C. It is an intermediate object while printing
D. None of these

Q9. Which of the following is a part of 5M's of management?

A. Mind
B. Men
C. Marketing
D. All of these

Q10. _____is used for the production of item that is too substantial or overwhelming to move?

A. Fixed layout
B. Functional layout
C. Cellular layout
D. Combination layout

Assignment 10

Q1. Head of the demonstrated machine can move in:

A. Y and Z direction
B. Z and X direction
C. X and Y direction
D. X, Y and Z direction

Q2. Precision level is higher in:

A. FDM
B. SLA
C. Both of these
D. None of these

Q3. Which is not a part of 5M's of management?

A. Mind
B. Money
C. Materials
D. Men

Q4. Flow shops that produce high volume, high standardized products prefer _____ kind of layout?

A. Functional
B. Cellular
C. Combination
D. Line

Q5. Factor influencing plant layout is/are:

A. Processing time
B. Product volume
C. Both of these
D. None of these

Q6. What is not an advantage of Process layout?

A. Flexibility
B. System protection
C. Motivation
D. Mass production

Q7. In which layout similar kind of processes are arranged in one section of a factory?

A. Functional layout
B. Line layout
C. Cellular layout

D. Combination layout

Q8. ‘Cura’ word is derived from:

A. Curry
B. Curing
C. Curious
D. None of these

Q9. In which layout the firm have the flexibility to deal with an assortment of routes and process requirements?

A. Line layout
B. Cellular layout
C. Functional layout
D. Combination layout

Q10. Product layout is also known as:

A. Functional layout
B. Line layout
C. Cellular layout
D. Combination layout

Assignment 11

Q1. In Tecnomatix software, Line represents:

A. Conveyor Belt
B. Block
C. Source
D. Machine

Q2. What is not true for digitalization?

A. Easy to store
B. Easy to profit
C. Easy to retrieve
D. Easy to edit

Q3. CAPP is:

A. Classical Automation Process Planning
B. Computer Aided Production Process
C. Computer Aided Process Planning
D. None of these

Q4. CAE is:

A. Computer Assisted Engineering
B. Computer Aided Evaluation
C. Computer Assisted Evaluation
D. Computer Aided Engineering

Q5. Which of the following is not classified as a secondary industry?

A. Manufacturing
B. Power generation
C. Construction
D. Service sector

Q6. Which of the following is true for CIM?

A. Improves the quality
B. Lowers the cost
C. Improves delivery time
D. All of these

Q7. Siemens PLM ERP does not include:

A. ANSYS
B. Tecnomatix
C. NX
D. Teamcenter

Q8. Which of these is a plant simulation software?

A. Arena
B. Alexa
C. Siri
D. All of these

Q9. MU stands for:

A. Manufacturing Unit
B. Mobile Unit
C. Marketing Unit
D. Memory Unit

Q10. MTTR is Minimum time to repair.

A. True
B. False

Assignment 11

Q1. Siemens PLM ERP includes:

A. ANSYS, Tecnomatix and Team up
B. NX, Tecnomatix and Team up
C. NX, Tecnomatix and Teamcenter
D. ANSYS, Tecnomatix and Teamcenter

Q2. Arena is confined to the plant simulation only.

A. True
B. False

Q3. FIFO is:

A. Fast In Fast Out
B. First In First Out
C. Fail In Fail Out
D. None of these

Q4. _____ have a plant simulation.

A. ANSYS
B. NX
C. Tecnomatix
D. None of these

Q5. Which is not a plant simulation software?

A. Arena
B. Alexa
C. Tecnomatix
D. None of these

Q6. Throughput per day in Tecnomatix is:

A. Total number of components that are purchased in a day
B. Total number of components that are sold in a day
C. Total number of components that are brought in a day
D. Total number of components that are manufactured in a day

Q7. GIGO is:

A. Garbage in garbage out
B. Grid in Grid out
C. Get in Get out
D. None of these

Q8. MU is:

A. Manufacturing unit
B. Mobile unit
C. Marketing unit
D. None of these

Q9. MTTR is:

A. Maximum Time To Repair
B. Minimum Time To Repair
C. Mean Time To Repair
D. Median Time To Repair

Q10. Mean Time To Failure (MTTF) is:

A. Maximum time between failure and maximum time to repair
B. Minimum time between failure and minimum time to repair
C. Maximum time between failure and minimum time to repair
D. Mean time between failure and mean time to repair

Assignment 12

Q1. Output of Reverse Engineering (RE) data acquisition is:

A. 2D cross-sectional image
B. 3D cross-sectional image
C. Line diagram
D. None of these

Q2. What is the correct sequence of benchmarking process?

A. What to benchmark, Understand the current performance, Plan, Study others, Learn from data, Use findings
B. Study others, Plan, What to benchmark, Understand the current performance, Learn from data, Use findings
C. Study others, Plan, Understand the current performance, What to benchmark, Learn from data, Use findings
D. Study others, Understand the current performance, Use findings, What to benchmark, Plan, Learn from data

Q3. Polygon models are not used for 3D graphing.

A. True
B. False

Q4. RE is an efficient approach to:

A. Minimizing the tool-workpiece lubrication
B. Promote CNC machining
C. Reduce the product development cycle
D. None of these

Q5. In mass customization, products are:

A. Standard shape with low product cost
B. Custom-made with low product cost
C. Standard shape with high product cost
D. Custom-made with high product cost

Q6. Sapphire probe tip Coordinate Measuring Machine is:

A. Non-contact type scanner
B. Contact type scanner
C. Both of these
D. None of these

Q7. Acquisition, Alignment, Mesh generation, Post processing, Simplification is a sequence of _____

A. Reverse engineering processes
B. Benchmarking
C. 3D scanning process
D. None of these

Q8. Which of the following is the right sequence of reverse engineering processes?

A. Processing of measured data, Digitizing of an object, Creation of CAD model, Prototype
B. Creation of CAD model, Processing of measured data, Digitizing of an object, Prototype
C. Prototype, Digitizing of an object, Processing of measured data, Creation of CAD model
D. None of these

Q9. There are three levels of benchmarking.

A. True
B. False

Q10. Operation that involves the enhancement and finishing of a mesh is known as:

A. Alignment
B. Mesh generation
C. Simplification
D. None of these

Assignment 12

Q1. What is the right sequence of 3D scanning processes?

A. Alignment, Acquisition, Mesh generation, Post processing, Simplification
B. Acquisition, Alignment, Mesh generation, Post processing, Simplification
C. Mesh generation, Acquisition, Alignment, Post processing, Simplification
D. Post processing, Simplification, Acquisition, Alignment, Mesh generation

Q2. Manual alignment is _____before using global alignment.

A. compulsory
B. advised
C. irreverent
D. None of these

Q3. Operations that involves the enhancement and finishing of a mesh is known as:

A. Post processing
B. Simplification
C. Alignment
D. Mesh generation

Q4. 'Reduce noise' filter is used:

A. To reduce mesh
B. To reduce noise
C. To smooth the surface
D. None of these

Q5. Mesh decimation is used:

A. To reduce mesh
B. To reduce noise
C. To smooth the surface
D. None of these

Q6. Reverse Engineering (RE) hardware is used for the:

A. RE manufacturing
B. RE machining
C. RE designing
D. RE data acquisition

Q7. Strategic benchmarking is done with a:

A. Long term perspective
B. Short term perspective
C. Averaged term perspective
D. None of these

Q8. Polygon models are used for:

A. Rapid prototyping
B. Laser milling
C. 3D graphing
D. All of these

Q9. Internal benchmarking is done:

A. With respect to the other company
B. Within company
C. Both of these
D. None of these

Q10. Performance metrics give _____ against which a client's own processes can be compared.

A. qualitative standard
B. absolute standard
C. relative standard
D. numerical standard